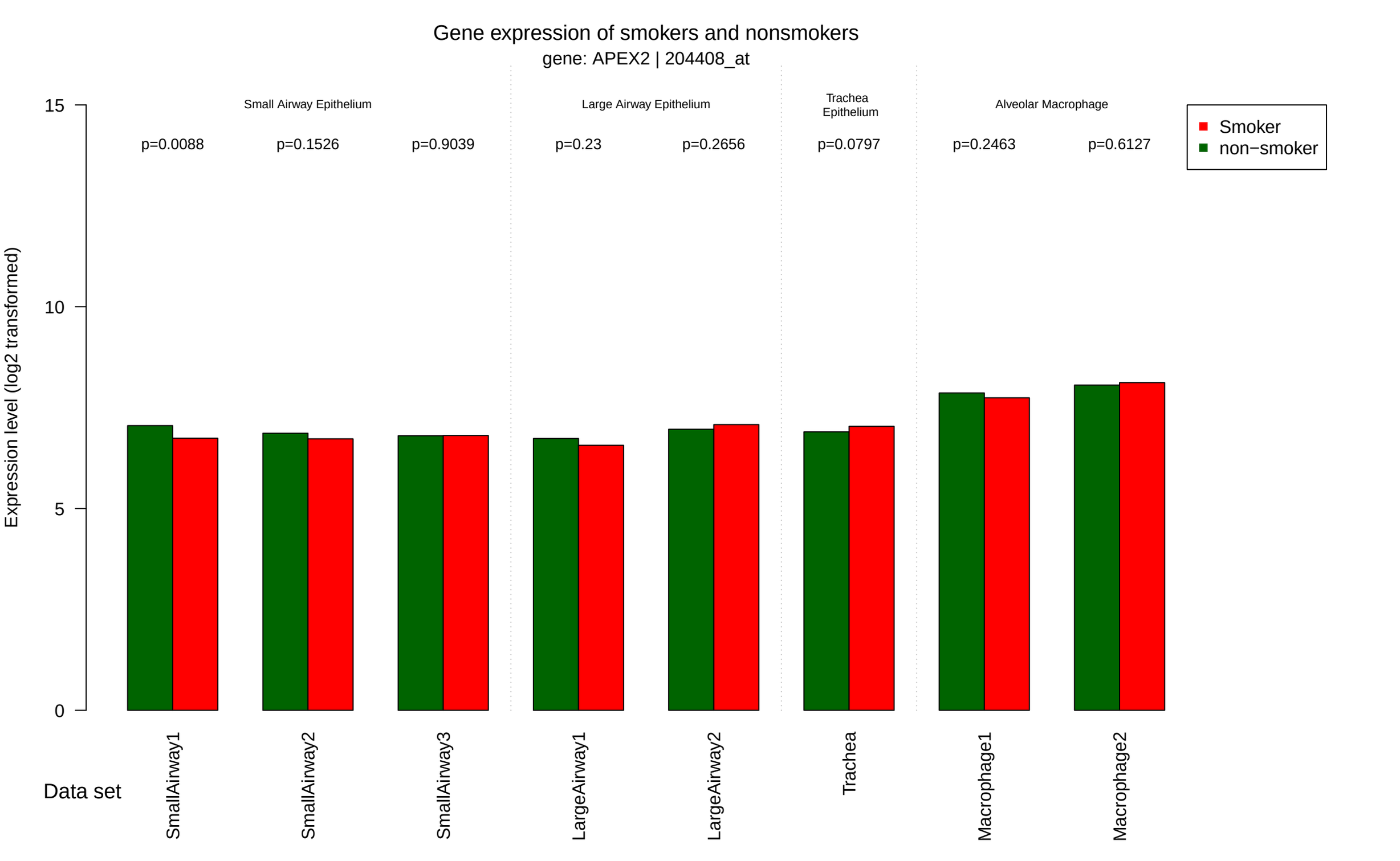
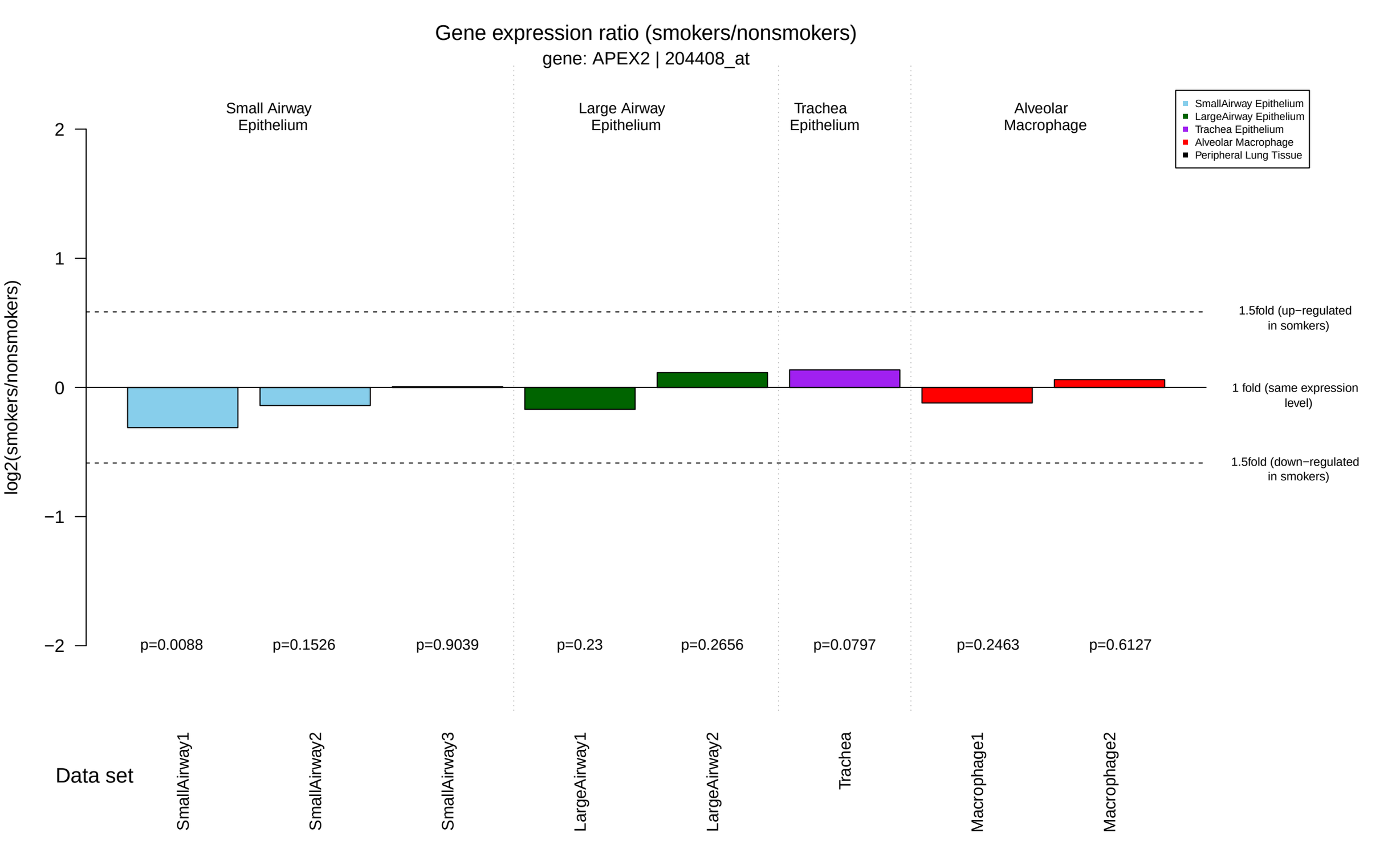
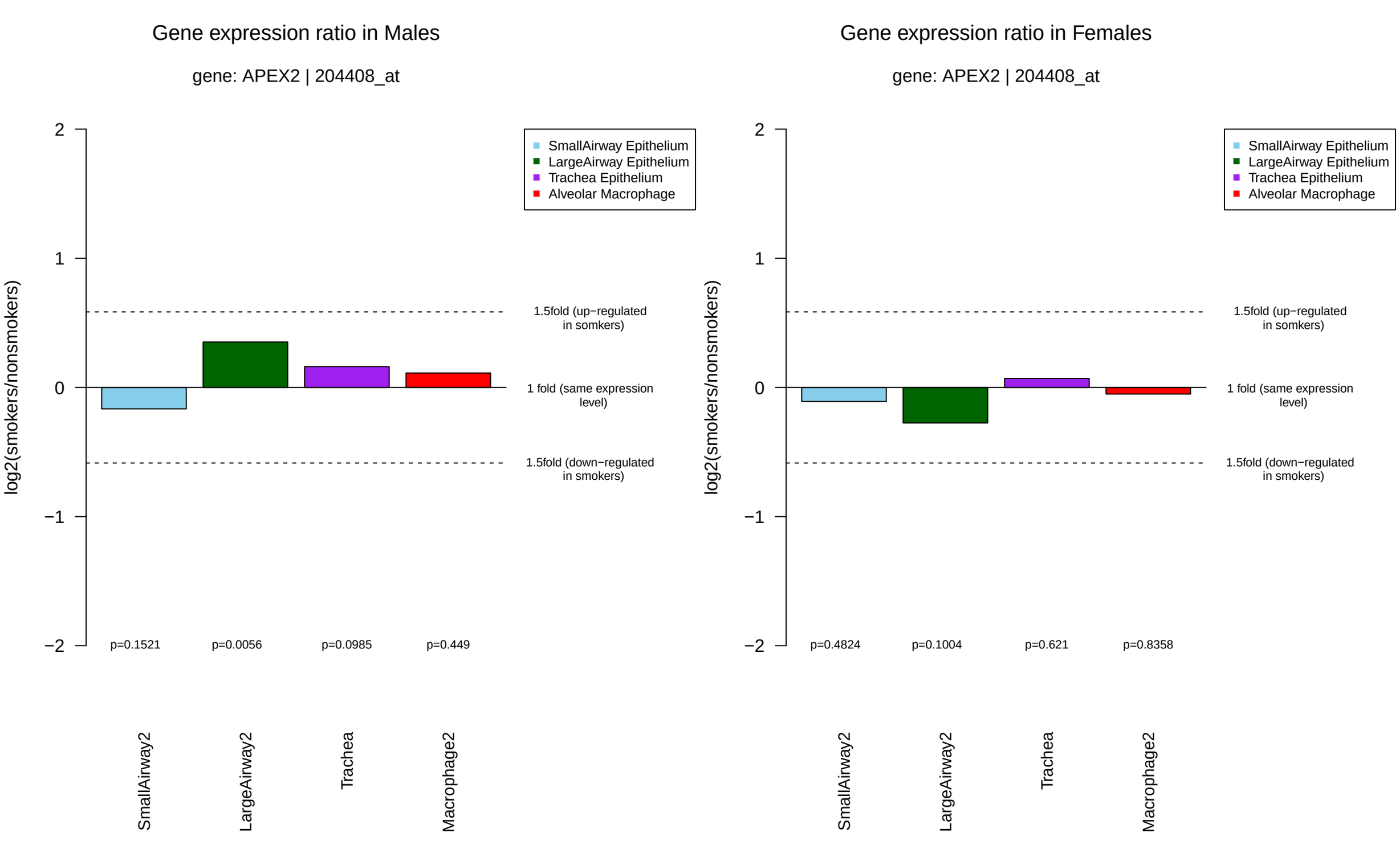
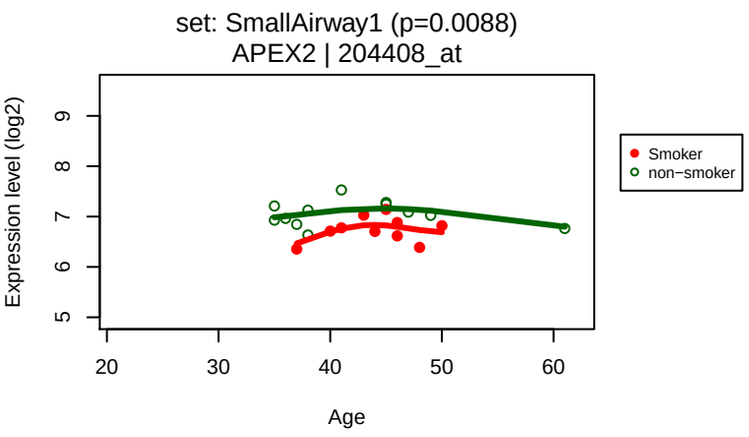
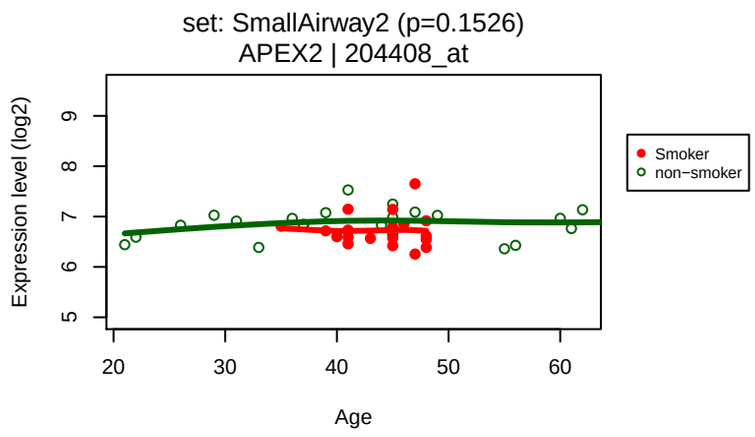
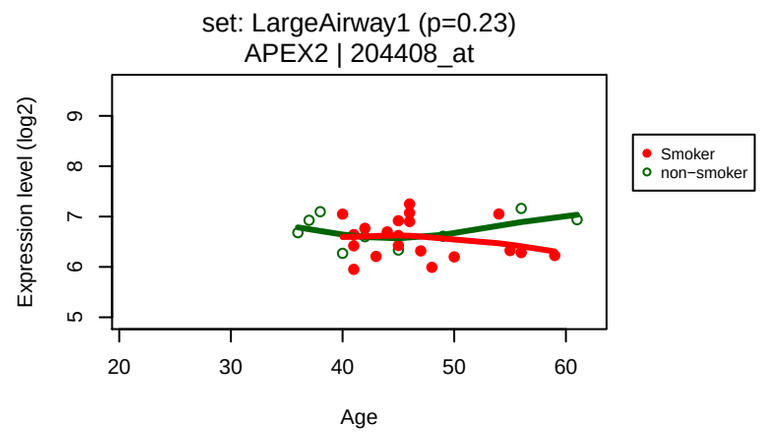

Gene expression of smokers and nonsmokers
gene: APEX2 | 204408_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0088
p=0.1526
p=0.9039
p=0.23
p=0.2656
p=0.0797
p=0.2463
p=0.6127
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

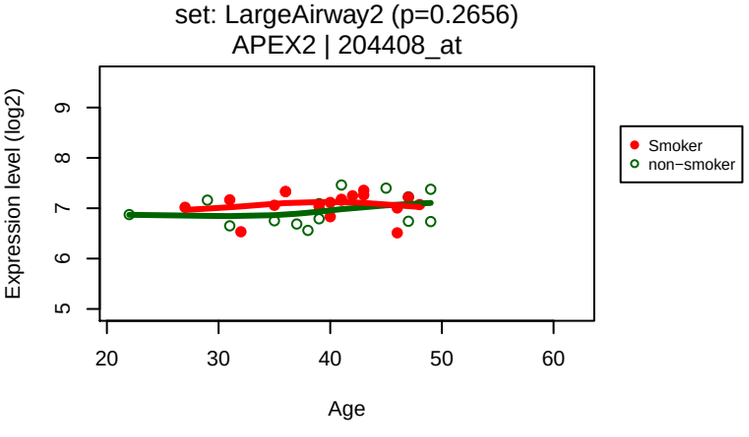

Gene expression ratio (smokers/nonsmokers)
gene: APEX2 | 204408_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0088
p=0.1526
p=0.9039
p=0.23
p=0.2656
p=0.0797
p=0.2463
p=0.6127
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

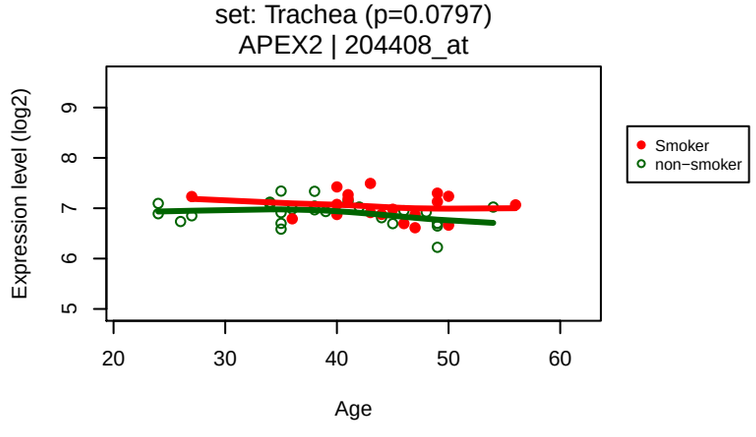

Gene expression ratio in Males
gene: APEX2 | 204408_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1521
p=0.0056
p=0.0985
p=0.449
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APEX2 | 204408_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4824
p=0.1004
p=0.621
p=0.8358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

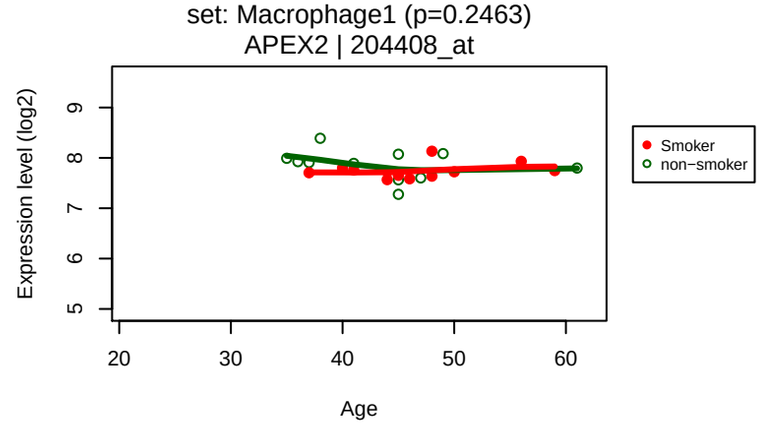

set: SmallAirway1 (p=0.0088)
APEX2 | 204408_at
Expression level (log2)
Age
Smoker
non−smoker

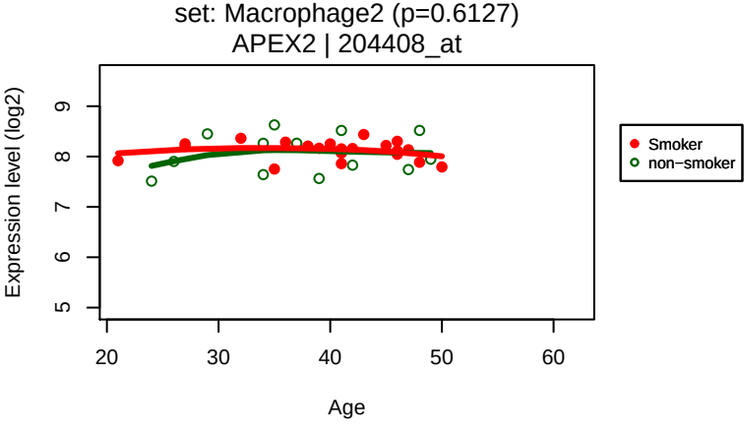

set: SmallAirway2 (p=0.1526)
APEX2 | 204408_at
Expression level (log2)
Age
Smoker
non−smoker

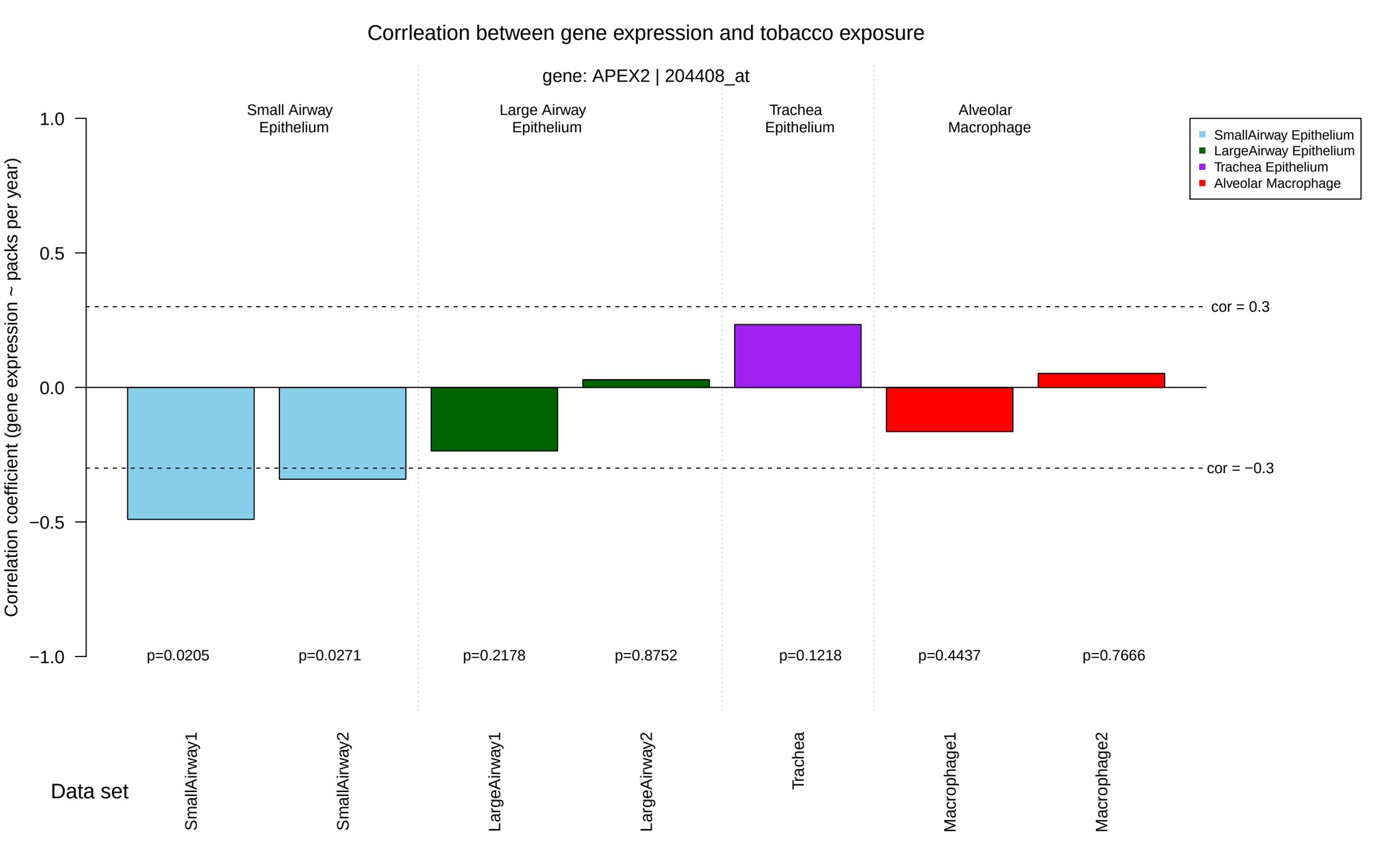

set: LargeAirway1 (p=0.23)
APEX2 | 204408_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2656)
APEX2 | 204408_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0797)
APEX2 | 204408_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2463)
APEX2 | 204408_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6127)
APEX2 | 204408_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APEX2 | 204408_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0205
p=0.0271
p=0.2178
p=0.8752
p=0.1218
p=0.4437
p=0.7666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2